

Databases

Igcse Computer Science

What is a database?

A **database** 数据库 is an organised store of data, kept so that it is easy to search, sort and update. At IGCSE you work with a **single-table database** 单表数据库—all the data is held in one table.

Records and fields

A database table is made of records and fields.

- A **record** 记录 is one row in the table—all the data about one thing (for example one student).
- A **field** 字段 is one column in the table—one piece of data that every record has (for example "First name").

StudentID	FirstName	DateOfBirth	FormClass	FeesPaid
1	Amy	14/03/2009	10A	TRUE
2	Ben	02/11/2008	10B	FALSE

Here each **row** is a record, and each **column** is a field.

Data types for fields

Each field stores one **data type** 数据类型. You choose the type that best fits the data.

Data type	Used for	Example
text/alphabetic 文本	letters, digits and symbols	"10A", "Amy"
character 字符	a single character	'M'
Boolean 布尔值	one of two values	TRUE / FALSE
integer 整数	a whole number	42
real 实数	a number with a decimal point	3.5
date/time 日期时间	a date or a time	14/03/2009

The primary key

A **primary key** 主键 is a field that holds a **unique** 唯一的 value for every record. No two records can have the same primary key, so it lets you pick out exactly one record.

In the table above, **StudentID** is a good primary key because every student has a different number. A field like **FormClass** would be a bad primary key, because many students share the same class.

Validation in a database

When data is put into a database, **validation** 验证 checks make sure it is sensible—for example a range check on an age field, or a presence check so a field is not left empty. (You saw these checks in topic 7.)

Structured Query Language (SQL)

Structured Query Language 结构化查询语言 (SQL) is a language used to **query** 查询 a database —to pick out the records you want. You must understand and complete SQL scripts.

SELECT, FROM and WHERE

- SELECT says **which fields** to show.
- FROM says **which table** to use.
- WHERE gives a **condition** 条件, so only matching records are shown.

```
SELECT FirstName, FormClass
FROM Student
WHERE FeesPaid = TRUE;
```

This shows the first name and class of every student who has paid the fees.

Use * to select **all** fields:

```
SELECT *
FROM Student
WHERE FormClass = '10A';
```

ORDER BY

ORDER BY sorts the results. Use ASC for **ascending** 升序 (smallest first, A→Z) or DESC for **descending** 降序 (largest first, Z→A).

```
SELECT FirstName, DateOfBirth
FROM Student
ORDER BY DateOfBirth ASC;
```

AND and OR

Join conditions with AND (both must be true) or OR (at least one must be true).

```
SELECT FirstName
FROM Student
WHERE FormClass = '10A' AND FeesPaid = FALSE;
```

SUM and COUNT

- SUM adds up the values in a number field.
- COUNT counts how many records match.

```
SELECT COUNT(StudentID)
FROM Student
WHERE FeesPaid = FALSE;
```

This counts how many students have not paid. SUM works the same way but adds a number field instead of counting rows.

Working out the output

To find the output of an SQL script, read it in this order:

1. FROM —which table;
2. WHERE —keep only the records that match the condition;
3. SELECT —show only the chosen fields;

4. `ORDER BY` —put the results in order.

Following these steps, you can write down exactly which rows and columns the query returns.